



Course Description	
COURSE NUMBER and NAME	DCS 404 Artificial Intelligence & Machine Learning
UNITS	3
LENGTH OF CLASS	8 Weeks
COURSE DESCRIPTION	This course introduces fundamentals of artificial intelligence and machine learning at the undergraduate level. The course introduces the theory underlying artificial intelligence and presents the creation of artificial intelligence algorithms. In succession, developing artificial intelligence software installments for various applications using the Python language is introduced. State of the art artificial intelligence models, including gaming models, are also presented. Additionally, the course emphasizes machine learning aspects of artificial intelligence, Q machine learning algorithms, and implementations are successively covered and presents deeper machine learning relevant statistical concepts and their implementation for development of learning schemes. Diverse applications of artificial intelligence and machine learning, including speech recognition and healthcare, are also examined.
REQUIRED TEXT	Nagy, Z. (2018). <i>Artificial intelligence and machine learning fundamentals</i> . Packt Publishing. ISBN: 9781789809206
INSTRUCTIONAL METHOD	Online / On-Campus

Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism (Rubric)	160	16%	1, 2, 3, 4, 5, 6, 7
Week 1 Assignment	70	7%	1
Week 2 Assignment	70	7%	1
Week 3 Assignment	100	10%	2, 3
Week 4 Assignment	100	10%	4



Assignments	Totals	Weight	CLOs
Week 5 Assignment	100	10%	5
Week 6 Assignment	100	10%	6
Week 7 Assignment	100	10%	7
Week 8 Final Assignment	200	20%	1, 2, 3, 4, 5, 6, 7
Total Points/Percentage	1000 Points	100%	

Course Policies

For Westcliff's course policies, please see the [Course Policies](#) document.

Discussion Requirements

For all discussions, the primary response is due by Wednesday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Saturday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines, and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, 1" margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents.

Participation Requirements



Students are required to attend each live class session either in person or virtually as stipulated in the course policies. Participation in the live class session is determined by actively engaging, answering or asking questions, providing comments, interacting in group activities, etc., as required by the instructor. Students who are unable to attend the live in-class or virtual sessions must follow the Virtual Class Session (VCS) submission requirements as stated in the Course Policies document.

[Writing Center](#)

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.



Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	PLOs
1. Demonstrate a basic understanding of Python.	1
2. Research and discuss the importance, principles, and field of AI.	1, 2
3. Demonstrate how to use Python to implement basic artificial intelligence for pathfinding and beating games.	1
4. Implement regression and classification exercises in Python applied to real-world problems.	4, 5
5. Perform predictive analysis in Python using decision trees and random forests.	4, 5
6. Perform clustering using the k-means and mean shift algorithms.	5
7. Research and discuss the fundamentals of deep learning via practical examples.	2



Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

- Videos & Tutorials:
 - Complete both Python Basics and Python Lists chapters in the [Introduction to Python](#) course provided for free by DataCamp.
 - Click [here](#) to go to the home page for this free course and then click on *Start Free Course*.
 - You will be required to register for a free account, if you don't have one already, to access this and other free content provided by DataCamp.

Week 1 Live Class Activity

To complete this and next week's assignments, students need to set up a free account with DataCamp and enroll in [Introduction to Python](#). For this activity, set up their accounts and enroll in the course.

Week 1 Assignment ([Rubric](#))

Submit a screenshot showing completion of both Python Basics and Python Lists chapters in DataCamp.



Week 2

Assignments to complete this week:

- Videos & Tutorials:
 - Complete both Functions and Packages and NumPy chapters in the *Introduction to Python* course provided for free by DataCamp.

Week 2 Live Class Activity

Python applications are more pervasive in the technology tools we use daily than one might think. For example, did you know that games such as The Sims, Battlefield 2, and Vega Strike all use Python? Where else is Python used?

In small groups, discuss different Python applications. Return and discuss as a class.

Week 2 Assignment ([Rubric](#))

Submit a screenshot showing completion of both Functions and Packages and NumPy chapters in DataCamp.



Week 3

Assignments to complete this week:

- Reading: *Intelligence and Machine Learning Fundamentals*
 - Chapter 1: Principles of AI
 - Chapter 2: AI with Search Techniques and Games
- Videos and Tutorials:
 - Read [*Introduction to AI and machine learning*](#) and watch and/or read the additional content.
 - Read [*Examples of what AI and ML can do*](#) and watch and/or read the additional content.
 - Course: Google Developers [*Introduction to ML Problem Framing*](#)

Week 3 Live Class Activity

In small groups depending on class size, go to [*Examples of what AI and ML can do*](#) and watch and/or read some of the examples of how AI and ML are used in the world. Discuss within the group some of the possible applications of AI and ML. Return and discuss as a class.

Week 3 Assignment ([Rubric](#))

Complete the [*Framing an ML problem*](#) and [*Formulating a solution*](#) exercises after completing the [*Introduction to Machine Learning Problem Framing*](#) course provided by Google Developers. At the end of each exercise there is a “Print Page” button. Click the button to save your responses as a PDF. Include these PDF files in your assignment submission for the week.



Week 4

Assignments to complete this week:

- Videos and Tutorials:
 - Familiarize yourself with [Google Colab](#) by watching [Get started with Google Colaboratory](#) video.
 - Complete [Data Preparation and Feature Engineering in ML](#) including the programming exercise at the end of the course.

Week 4 Live Class

This week and in the weeks ahead, you will be working in Python using [Google Colab](#). For this activity, access Google [Colab](#), and set up your access. Familiarize yourself with this Python coding environment by clicking on available content. Watch the [Get started with Google Colaboratory](#) video during class and ensure that all students have access and are set up to use these tools.

Week 4 Assignment ([Rubric](#))

Complete the [Text Classification with Movie Reviews ML](#) tutorial with TensorFlow using [Google Colab](#). Print the notebook and save it as a PDF. Submit the PDF for the assignment this week.



Week 5

Assignments to complete this week:

- Reading: *Intelligence and machine learning fundamentals*
 - Chapter 3: Regression
 - Chapter 4: Classification

Week 5 Live Class

As a class, watch [The 7 steps of machine learning](#) video. Discuss the concepts and ideas addressed during the video as a class. Break into small groups depending on class size and discuss the applications of ML in business settings. Be prepared to discuss your ideas with the live class.

Week 5 Assignment ([Rubric](#))

Complete the following exercises and activities at the end of Chapter 3:

- Activity 5: Predicting Population
 - Use the guide for this activity starting on page 269 of the textbook.
- Activity 6: Stock Price Prediction with Quadratic and Cubic Linear Polynomial Regression with Multiple Variables
 - Use the guide for this activity starting on page 271 of the textbook.

Complete the following exercises and activities at the end of Chapter 4:

- Exercise 10: Loading Datasets
- Exercise 11: Pre-Processing Data
- Activity 7: Preparing Credit Data for Classification
 - Use the guide for this activity starting on page 276 of the textbook.
- Exercise 12: Illustrating the K-nearest Neighbor Classifier Algorithm
- Exercise 13: k-nearest Neighbor Classification in scikit-learn
- Exercise 14: Prediction with the k-nearest neighbor classifier
- Activity 8: Increasing the Accuracy of Credit Scoring
 - Use the guide for this activity starting on page 280 of the textbook.
- Activity 9: Support Vector Machine Optimization in scikit-learn
 - Use the guide for this activity starting on page 280 of the textbook.

Each exercise and activity will have you using notebooks in [Google Colab](#) to practice using Python code in different machine learning scenarios. Once you've completed each exercise, you will print and save your notebook as a PDF and submit it as part of your assignment for this week.



Week 6

Assignments to complete this week:

- Reading: *Intelligence and machine learning fundamentals*
 - Chapter 5: Using Trees for Predictive Analysis
- Article:
 - [An Introduction To Decision Trees and Predictive Analytics](#)
- Videos and Tutorials:
 - Complete the [Decision Forests](#) course provided by Google Developers.

Week 6 Live Class Activity

Review the [Introduction To Decision Trees and Predictive Analytics](#) (2021) article in small groups depending on class size, and discuss how decision trees and predictive analytics are used in data science to support strategic decision making. Be prepared to discuss your ideas with the live class.

Week 6 Assignment ([Rubric](#))

Throughout the [Decision Forests](#) course, there will be short quizzes to check your understanding. After completing each quiz, print and save the page/results as a PDF. Compile all printed pages and submit them for your assignment this week.



Week 7

Assignments to complete this week:

- Reading: *Intelligence and machine learning fundamentals*
 - Chapter 6: Clustering

Week 7 Live Class Activities

Activity 1

As you'll learn in completing the [Clustering](#) course this week, clustering is used in a variety of industries. Some examples include:

- Market segmentation
- Social network analysis
- Search result grouping
- Medical imaging
- Image segmentation
- Anomaly detection

In small groups depending on class size, pick one of these examples and discuss how clustering is applied. Return and discuss as a class.

Activity 2

Review the Week 8 Final Assignment and allow class time for students to begin preparation.

Week 7 Assignment ([Rubric](#))

Complete the [Clustering with k-means: Programming Exercise](#) at the end of the [Clustering](#) course. This programming exercise has two parts:

- [Clustering with a Manual Similarity Measure](#)
- [Clustering with a Supervised Similarity Measure](#)

After completing each exercise in [Google Colab](#), print the notebook and save it as a PDF. Submit both PDF documents for your assignment this week.

Begin preparing for your Week 8 Final Assignment, which will be presented during your live class session.

Week 8

Assignments to complete this week:

- Reading: *Intelligence and machine learning fundamentals*
 - Chapter 7: Deep Learning with Neural Networks

Week 8 Live Class Activity

Students will present their final presentations to the class. Students should give attention to each presentation and consider questions that they might ask.

Week 8 Final Assignment ([Rubric](#))

Review the information you have studied during the first 7 weeks of this course. Select 3 key items you have learned and develop a brief PowerPoint presentation of 5-7 slides to be delivered during the Week 8 live class session.